Syllabus

Learning Outcomes

The course teaches the basic techniques and concepts relevant to Space Traffic Management (STM) and Space Situational Awareness (SSA) in the near-Earth realm from a rigorous engineering perspective.

The class starts with learning about measurement geometries, then moves on to the non-linear orbital motion and astrodynamics. The focus is on orbit determination and orbit improvement to detect new objects and maintain custody of them. Lastly, the class focuses on determining the probability of collision of two (or more) resident space objects.

Upon completion of the class, the students are equipped with a fundamental toolset to investigate their own STM or SSA problems. Specific learning outcomes are: Signals: Measurement simulation and processing. Dynamics: orbit determination, propagation in the presence of uncertainty in the non-linear regime. Presentation and communication skills: Presentation of formulation of technical content, critical evaluation of published technical content.

The class is open to senior undergraduate students and counts as Dynamics and Controls area for them.

Required Course Materials

No book is required; the course script will be made available on Brightspace.

Optional books for reference that cover parts of the course materials are (selection):

- David Vallado, Fundamentals of Astrodynamics and Applications, Microcosm Press, 2013
- Oliver Montenbruck, Satellite Orbits - Models, Methods and Applications, Springer, 2000
- Bob Schutz, George Born, Byron Tapley, Statistical Orbit Determination, Academic Press, 2004

The use of and coding in Matlab is required for this course; a student edition is sufficient; helpful Matlab packages are the statistics and image processing package.

Topics to be covered:

1. Quick recap on rotations and probability
2. Sensors, CCD response in astrometric observations
3. Influence of the optics in astrometric measurements
4. Coordinate systems space fixed: right ascension, declination, geocentrioc, topocentric
5. Coordinate systems Earth fixed: elevation, azimuth, aberration
6. Coordinate systems: time systems, hour angle computation
7. Coordinate systems: J2000, nutation and precession models simple and complex
8. Orbit propagation: spherical expansion of the gravity field and pines implementation

9. Orbit propagation: third body effects
10. Orbit propagation: SRP and drag
11. Uncertainty Propagation
12. Probability of collision between two space objects and time of closes approach.
13. Initial orbit determination: admissible regions
14. IOD: Admissible regions: connection of two regions
15. First orbit improvement: least squares, introduction linear least squares
16. First orbit improvement: least squares, non-linear least squares
17. First orbit improvement: covariance discussion
18. 2nd orbit improvement: Kalman filter as a test case
19. 2nd orbit improvement: Extended Kalman filter in the orbit problem
20. Orbit improvement and orbit propagation
21. Two line elements and SGP4
22. Space object characterization using optical measurements

Grading

5% brightspace quizzes and questions
40% homework
45% midterm and final exam
10% final project

Letter grades are assigned at the end of the semester ranging from A+ to F.

Copyright, Sources, AI, and Large Language Models

All course materials are copyright protected. Students are free to use them as part of the class. Distribution is only permitted with instructor approval. Uploading materials on webshare sites (e.g., CourseHero) is explicitly prohibited.

The use of all sources used in the homework, and for quizzes have to be explicitly declared. It is the expectation that all homework you hand in is your own work.

AI and Large Language Models (e.g., ChatGPT, Mistral AI, DeepSeek, OpenAI, Claude, Llama, Grok, Falcon, BERT, PaLM, Gemini, T5, Turing NLG....) is not permitted unless explicitly stated. All parts of LLMs need to be explicitly declared as sources and all parts have to be explicitly marked as direct quotations. You are free to use spell checkers or grammar tools like Grammarly, or dictionary tools like LEO to enhance your English language writing.

Brightspace Quizzes and Questions

Weekly a small single question quiz is posted and to be answered on Brightspace. Each student is asked to also provide a question and the correct answer to them via gradescope once in the course of the semester. The submitted questions are intended to be used as quizzes throughout the semester, however, they might be altered or modified as seen fit. Note: When suggesting questions, it is all

about asking something that has been super interesting to you, or difficult, or confusing to yourself; the answer that you provide to your own question is not graded for correctness. You may also state that you are unsure and just give your best attempt at answering it. The schedule of questions to be delivered is posted on Brightspace during the first week of class.

Homework

Homework is to be handed in as a pdf file, via gradescope.

Where it applies, the code is to be embedded in the pdf in the relevant parts and handed in a runable form via a zip file (relevant instructions are provided on the homework problem sheets). The lowest homework grade will be dropped with one exception: One of the homeworks will be a literature review; this homework cannot be the one you are dropping.

Homework Due Dates

Homework is expected to be submitted by the time indicated on the homework sheet. When submitted within 2h of the indicated due date, no points are deducted. When submitting 24h or more after the due date, your homework is still graded, but zero points are awarded. In the time window between 2 and 24h after the due date the maximum obtainable points are gradually reduced commensurate to the time passed since the due date. Each student has two 24h extensions to homeworks throughout the semester. Students may use those extensions at any time and are automatically applied to the first two times homework is submitted past the 2h courtesy period and within 24h of the original due date and time (excluding the 2h courtesy period). NOTE: Always go off the homework sheet and not the gradescope time; to accomodate the variable 24h period, it might appear like there is an extension on gradescope, although you cannot take an extra 24h on more than two assignments in the semester.

Code guidelines: Homework and Code Example

The code has to follow a pep8 coding style. Pep8 is a format originally designed for Python. The external document published on Brightspace outlines the formatting requirements and best practices and how they are adapted to the Matlab environment. Every third line has to be a comment.

Final Project

The class has a very small final project. Design and code a playable single or multi-player game using the materials learned in class. Write a small description document on how to play it (no minimum pages, no page limit) and deliver the code to actually play it. If you want to share your game with your peers, you can do so on Piazza, but it is not mandatory.

Exams

This class has midterm and the final exam. In terms of grading, if the final exam has a higher grade than the midterm, the final exam grade is taken as your exam grade. If the final exam grade is lower

than the grade of the midterm, the average of the two are used as your exam grade.

Boilermaker Pledge and University Resources

As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue.

Nondiscrimination Statement
Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person and encourages each individual to strive to reach their own potentia in pursuit of its goal of academic excellence. Act accordingly!

Mental Health
If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS). TAO is available to all students at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website. If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

Academic Integrity
Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breeches of this value by either emailing `integrity@purdue.edu` or by calling 765-494-8778. While information may be submitted anonymously, the more information that is submitted provides the greatest opportunity for the university to investigate the concern.

Emergency Preparedness and Readiness
See the separate document posted on Brightspace in the Emergency Preparedness and Readiness folder.